

aggressive in an order even in times of illiquidity and adverse price movement if doing so would result in a significant reduction of residual basket risk. The biggest difference between single stock and basket algorithms is that the basket algorithm will manage cost and total basket risk (correlation and covariance) whereas the single stock algorithm will seek to manage the cost and individual risk of the stock. Important basket trading constraints include cash balancing, self-financing, minimum and maximum participation rate.

Risk Aversion Parameter: The meaning of the risk aversion parameter used across the different brokers will vary. First, the optimization technique is not constant. For example, some parties will optimize the trade-off between cost and variance since it fits a straightforward quadratic optimization formulation. Others optimize based on the trade-off between cost and standard deviation (square root of variance) which results in a non-linear optimization formulation. Second, the definition of the risk aversion parameter, usually denoted by λ , varies. Some brokers specify $\lambda = d\text{Cost}/d\text{Risk}$ where $\lambda > 0$. Some map λ to be between 0 and 1 (0 = most passive and 1 = most aggressive), and still others map risk aversion to be between 1 and 3 or 1 and 10. Thus selecting a value of $\lambda = 1$ could mean the most aggressive strategy, the most passive strategy, or somewhere in the middle. Still others use a qualitative measure such as passive, medium, aggressive, etc., rather than a specified value of λ . Investors need to discuss the meaning of the risk aversion parameter with their providers in order to determine how it should be specified in the optimization process in order to ensure consistency across trading goal and investment objective.

Black Box Algorithms: The family of black box trading algorithms are commonly referred to as profit and loss algorithms and/or robo trading algorithms. These include all algorithms that make investment decisions based on market signals and execute decisions in the marketplace. Unlike the implementation algorithms that are tasked with liquidating a pre-determined position within some specified guidelines or rules, the black box algorithms monitor market events, prices, trading quantities, etc., for a profiting opportunity search. Once profiting opportunity appears in the market, the algorithm instantaneously buys/sells the shares. Many black box algorithms have time horizons varying from seconds to minutes, and some longer time horizons run from hours to days. While many investors use blackbox algorithms, they are still primarily tools of the quants, and especially when it comes to high frequency trading. Some black box trading algorithms are pairs-trading, auto market making, and statistical